

Does Learning a Wavelet Help?

Perfect-Reconstruction Constraints Enable Learnable Wavelets for Low-Dose CT Denoising

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Abstract—Making the wavelet transform learnable is a natural extension of wavelet-domain image denoising. We show that for low-dose CT, this extension fails by default: an unconstrained learnable analysis–synthesis pair drifts away from invertibility during training, and the resulting network performs no better than a fixed Haar basis. Our design resolves a 0.136 dB effect at $n = 5$ seeds, and the unconstrained–fixed gap is 0.025 dB — $5.6\times$ below that floor — so we state the result as a *bound* rather than an absence: any real difference is smaller than 0.14 dB, roughly *half* the benefit that perfect reconstruction later delivers. We trace the failure to the transform’s reconstruction error, which reaches 11–18% after training. We show this is causal, not merely correlated: injecting a controlled error of the same magnitude into the perfect-reconstruction arm degrades it to the fixed-Haar level. Imposing perfect reconstruction *by construction* — parameterizing the wavelet as a lifting scheme whose inverse shares the forward filters — makes learning pay off: +0.30 dB over unconstrained learning ($p = 0.0005$, $d_z = 4.67$) and +0.27 dB over fixed Haar ($p = 0.0004$, $d_z = 4.78$) — large effects, not marginal ones — consistent across two data splits. The resulting network has 2,159 parameters and reaches 31.97 dB on AAPM-Mayo 2016, within 1.0 dB of a RED-CNN baseline trained under the identical protocol at $857\times$ fewer parameters.

Two further results locate the effect. First, the transform learns *little*: from any numerically intact starting point, training converges to a narrow band of drift from Haar and, starting at Haar, moves away — the optimum is slightly off Haar, which is why adaptation buys so little here. We also reject the natural explanation that the heads simply substitute for the transform: the PR advantage does not shrink as head capacity grows by two orders of magnitude. Second, the failure mode is a cliff, not a slope — once the forward transform’s round-trip error exceeds the signal, the gradient carries no information and training cannot recover (drift 4.0 stays at 4.000 and 3.29 dB, where drift 2.0 recovers fully). Being inside the numerical envelope is therefore binary, which is why bounding is necessary — and why the bound’s value matters more than expected: our default costs +0.10 to +0.15 dB, about half the entire PR effect.

Index Terms—low-dose CT, image denoising, learnable wavelet, lifting scheme, perfect reconstruction, invertibility

I. INTRODUCTION

Low-dose CT (LDCT) reduces radiation exposure at the cost of increased noise. Deep denoising networks such as RED-CNN [1] are effective but parameter-heavy ($\sim 10^6$ parameters), which limits deployment on the resource-constrained workstations common in smaller clinical sites. Compact models are therefore attractive.

Wavelet-domain processing is a classical way to separate noise from structure, and recent work has made the wavelet itself **learnable** to adapt to a specific modality and dose [17], [18], [19], [20], [21], [22]. This paper asks a simple question:

Does making the wavelet learnable actually help?

Recent invertible-wavelet denoisers [19], [20] build perfect reconstruction (PR) into their architectures. Whether that structure is *what makes learning pay* has never been tested: prior work **assumes** invertibility, so it cannot observe its absence. Our answer, from a controlled three-arm study that makes invertibility **optional** while holding capacity, split, schedule and evaluation fixed, is: **learning pays only if the transform is kept invertible by construction**.

Parameterizing the wavelet as a **lifting scheme** [23], [24] — the inverse *reuses* the forward prediction/update filters rather than being a second set of parameters — makes learning pay off: **+0.30 dB over unconstrained learning and +0.27 dB over fixed Haar**, at large effect size ($d_z \approx 4.7$), replicated on a second split and on SSIM.

The inverted case is equally informative. An unconstrained learnable analysis–synthesis pair performs **statistically indistinguishably from a fixed Haar basis**. We are careful how we state this: the design resolves a 0.136 dB effect at $n = 5$ seeds, whereas the unconstrained–fixed gap is 0.025 dB — $5.6\times$ **below that floor**. So we report a **bound**, not an absence: any genuine difference is smaller than **0.14 dB**, roughly half of what PR delivers. A weak experiment could not have resolved the effect that PR produces.

Why it fails is the substance of the paper. After training, the unconstrained pair loses **11–18%** of the signal in a forward–backward pass, against **$1.6e-07$** for both invertible arms. We show this is **causal, not merely correlated**: holding architecture, capacity and protocol fixed, we inject a controlled round-trip error into the PR arm, and it degrades **monotonically** to the fixed-Haar level (8.7% error $\rightarrow$ 30.5500 dB, against 30.5591 dB for an arm that never learned anything; $p \leq 0.007$ at every dose). Prior invertible-wavelet denoisers cannot observe this state, because their designs make it unreachable.

Contributions.

- 1) **A missing control**, and the negative result it exposes: with capacity and protocol held fixed, an unconstrained

learnable wavelet gives no benefit over fixed Haar — reported as a **bound of 0.14 dB**, not as an absence.

- 2) **A causal mechanism**, established by intervention rather than correlation: a round-trip error of the same magnitude is *sufficient* to destroy the benefit, monotonically and significantly ($n = 5$ seeds, $p \leq 0.007$ at every dose). Prior invertible-wavelet denoisers cannot observe this state.
- 3) **A numerical envelope, and what happens outside it**. Structural PR does not enforce itself in float32. We characterize the boundary in two ways. By a **controlled sweep of tap magnitude**, the round-trip error grows from $3.6\text{e-}07$ at $|\text{taps}| \approx 1.0$ to $5.2\text{e+}00$ at ≈ 5.3 (*a controlled analysis of the parameter regime, not a training-drift measurement*). And by initialising the taps at a known distance from Haar, we show that crossing this boundary is a **cliff**: at drift 4.0 the forward transform emits noise, the gradient carries no information, and training stays exactly where it started (final drift 4.000, 3.29 dB), whereas drift 2.0 recovers fully. Being inside the envelope is therefore not a matter of degree — which is why we bound the taps, and why the bound’s value matters more than we expected (Section V-D).
- 4) **What the transform actually learns**. From any numerically intact starting point, training converges to a **narrow band of drift from Haar** ($\approx 0.65\text{--}0.78$); starting at Haar, it moves away. The optimum is therefore slightly off Haar, which is the concrete reason “learning” buys so little here. We also test — and reject — the natural hypothesis that the heads simply substitute for the transform: the PR advantage does not shrink as head capacity grows by two orders of magnitude.
- 5) **A 2,159-parameter network** that comes within 1.0 dB of a same-protocol RED-CNN baseline at $857\times$ **fewer parameters**.

II. RELATED WORK

A. Low-dose CT denoising

Convolutional encoder–decoders such as RED-CNN [1] established the field, followed by adversarial [2], [3] and, more recently, transformer- and diffusion-based methods [4], [5], [6], [7], [8], [9], [10], [11]. These improve fidelity but at $10^5\text{--}10^7$ parameters. A parallel line targets efficiency instead: MLAR-UNet [13], CT-Mamba [15], AMFA-Net [16], FMD-Net [14] and MobileMamba-UNet [12] all report competitive results at far lower cost, the last of these also operating in a wavelet domain. We share the efficiency goal but ask a different question — not *how small can a denoiser be*, but *does making the transform itself learnable help at all*.

B. Wavelet-domain and multiresolution denoising

Wavelet shrinkage [27], [28] separates noise from structure by thresholding subband coefficients. Multi-level wavelet CNNs [17], [18] fold this structure back into deep networks, and recent LDCT methods pair wavelets with diffusion [11] or

state-space models [12]. In all of these the transform is a **fixed** operator; the learnable capacity sits in the network around it.

C. Learnable transforms and invertibility

A separate line makes the transform itself learnable. LINN [19] learns lifting prediction/update steps around a *fixed* undecimated Haar transform; WINNet [20] additionally learns the splitting operator, constrained to be orthogonal via a Cayley transform so that PR is preserved by construction. Both therefore **build invertibility into the architecture**. Classical filter-bank theory [25], [26] supplies the PR conditions, the lifting scheme [23], [24] is the standard construction that satisfies them by design, and orthogonality has separately been used in CNNs to condition optimisation [29].

D. The closest prior result, engaged directly

WINNet’s Table II ablation compares decimated and undecimated Haar, DCT of several sizes, and learned Cayley operators on natural-image AWGN denoising, and reports that the **learned** operator performs *similarly to a fixed DCT* — whereupon they default to the fixed one. This superficially resembles our negative result, so we address it here rather than leave it for a reviewer. Two differences matter. First, their learned operator is constrained to be orthogonal, so PR holds throughout: their comparison is between two *invertible* transforms, whereas ours is between invertible and **non**-invertible ones — the failure we diagnose is not reachable in their design, and neither paper reports a round-trip measurement anywhere. Second, theirs is a default-picking ablation without significance testing; ours is a five-seed paired comparison with confidence intervals.

E. Our position

We propose neither a new wavelet nor a new architecture — the lifting construction in Section III-B is LINN’s and WINNet’s, and WINNet already learns the filter taps while preserving PR. What is new is the **missing control**. Prior invertible-wavelet denoisers *assume* structural invertibility and never test its absence, and so cannot report what happens without it. Making a learnable transform’s invertibility *optional*, holding capacity and training fixed, and measuring the consequence — and the six-order-of-magnitude round-trip gap that explains it — is the contribution. Section V-B returns to the residual tension with WINNet’s ablation.

III. METHOD

A. Overview

The network operates in the wavelet domain:

$$\begin{aligned} x &\xrightarrow{\text{PR-LWT}} \{\text{band}_b\}_{b=1}^7 \xrightarrow{\text{heads}} \{\text{band}_b - \delta_b\} \\ &\xrightarrow{\text{PR-LWT}^{-1}} \hat{x} \end{aligned} \quad (1)$$

where x is the noisy input and $\delta_b = \text{head}_b(\text{band}_b)$ is a per-subband residual prediction. For 2 levels this gives seven subbands $\{\text{LL}_2, \text{LH}_2, \text{HL}_2, \text{HH}_2, \text{LH}_1, \text{HL}_1, \text{HH}_1\}$.

B. PR-LWT: a perfectly-reconstructible learnable wavelet

We use the **lifting scheme**. In 1-D, with even/odd splitting,

$$\begin{aligned} \text{Analysis: } d &= x_o - P(x_e), & s &= x_e + U(d) \\ \text{Synthesis: } x_e &= s - U(d), & x_o &= d + P(x_e) \end{aligned} \quad (2)$$

Crucially, **synthesis reuses the same P and U** as analysis. Invertibility is therefore a property of the *structure*, not of the learned values: for any P, U , $\text{reconstruct}(\text{decompose}(x)) = x$ up to floating point.

The 2-D transform is separable (lifting along height, then width). We initialize $P = [0, 1, 0]$ (identity prediction) and $U = [0, 0.5, 0]$, plus a $\sqrt{2}$ output scaling, so that the initialization is **exactly orthonormal Haar** (verified to 2×10^{-7} on all four subbands).

Parameter bounding is necessary, not optional. We parameterize

$$P = P_{\text{init}} + 0.5 \tanh(\theta_P), \quad U = U_{\text{init}} + 0.5 \tanh(\theta_U) \quad (3)$$

Without this, the taps drift to $|P| \approx 5$, at which point the float32 round-trip error degrades from 3.6×10^{-7} to 5.2×10^0 . Perfect reconstruction holds unconditionally in exact arithmetic; bounding is what makes it *usable* in float32.

Parameter count: 2 levels $\times$ 2 directions $\times$ 2 filters $\times$ 3 taps = **24**.

C. Subband denoising heads

Each subband is processed by a lightweight head ($\text{conv}_{3 \times 3} \rightarrow \text{ReLU} \rightarrow \text{conv}_{3 \times 3}$, 16 internal channels) that predicts a **residual** (signed, no final activation). Seven heads contribute 2,135 parameters; the complete network has **2,159**.

D. Training

Loss: L_1 between prediction and ground truth. Adam, lr 10^{-3} , 30 epochs, 128×128 random patches, batch size 8.

IV. EXPERIMENTAL SETUP

A. Data

AAPM-Mayo 2016 Low Dose CT Grand Challenge [30], [31], 3 mm B30 paired subset: 10 patients, 2,378 quarter-dose / full-dose 512×512 slice pairs. Slices are paired by `ImagePositionPatient` [2] (z position); pairing by filename is **not** possible because the slice-index field is identical on both sides.

B. Evaluation protocol

AAPM never defined a PSNR protocol (its official metric was radiologist reading), and at least five mutually incompatible conventions coexist on this dataset. We adopt the **RED-CNN / CTformer lineage** convention, used verbatim by six public repositories:

preprocess: $x = (\text{HU} + 1024) / 4096$, no clipping.

evaluate: denormalize to HU, clip both prediction and reference to $[-160, 240]$, and take $\text{PSNR} = 10 \log_{10}(400^2 / \text{MSE}_{\text{HU}})$, i.e. `data_range = 400`; SSIM per SSIMu/RED-CNN.

aggregate: per-slice metrics, then mean.

The identity baseline under this protocol is L506 29.2489, L067 26.5576, combined 27.8630 dB. **We do not claim**

TABLE I

PATIENT-LEVEL SPLITS. S2 ALIGNS WITH THE CONVENTION USED BY PUBLISHED WORK SO THAT NUMBERS ARE DIRECTLY COMPARABLE.

Split	Train	Val	Test
S1 (main)	7 pat. (1,600)	L291	L506+L067 (435)
S2 (lit.)	8 pat. (1,943)	L067	L506 only (211)

TABLE II

IDENTITY BASELINE, ALL TEN PATIENTS. THE SPREAD (25.50–29.25 dB, A 3.75 dB RANGE) IS LARGER THAN ANY EFFECT WE REPORT, SO **ALL RESULTS ARE REPORTED PER PATIENT**.

Patient	n	PSNR	SSIM	Appears as
L143	234	25.4991	0.7643	train
L310	214	26.2577	0.7273	train
L067	224	26.5576	0.7987	S1 test / S2 val
L109	128	26.6052	0.8165	train
L096	330	26.9097	0.7853	train
L291	343	26.9805	0.8088	S1 val / S2 train
L333	244	27.2508	0.8305	train
L192	240	28.2896	0.8344	train
L286	210	29.1479	0.8103	train
L506	211	29.2489	0.8759	test (both)
<i>S1 test, combined</i>	435	27.8630	0.8361	
<i>S2 test (L506)</i>	211	29.2489	0.8759	

these as an independent confirmation, and we caution against reading cross-paper PSNR on this dataset at face value: independent re-analyses of the same cohort differ by up to **~ 2 dB** (a third-party report gives 27.24 dB on L506 against our 29.2489), and the Eulig et al. benchmark places its identity-equivalent reference 0.8 dB away. Every claim below is therefore made against **same-protocol baselines** rather than against quoted numbers.

C. Splits

Two patient-level splits are used to show robustness (Table I).

D. The floor: doing nothing

Identity (output = input) is the floor every model must beat. All ten patients, sorted by PSNR, are given in Table II.

V. EXPERIMENTS

A. Main results

Main results are in Table III.

The baselines marked *ours* were trained by us under the identical protocol — same splits, same patch size, batch size, optimiser, schedule and evaluation — so the comparison no longer rests on matching a published setup. Two observations follow.

First, an **independent check of our protocol**: on the literature-aligned split S2 our RED-CNN reaches **32.9471 dB on L506**, within **0.02 dB** of the independently published 32.93 dB for the same patient [1]. Our implementation and our evaluation convention therefore reproduce the reference result rather than merely resembling it.

TABLE III

MAIN RESULTS. ROWS MARKED *ours*, *same split* WERE TRAINED BY US UNDER THE IDENTICAL PROTOCOL. OUR RED-CNN REACHES 32.9471 dB ON L506, WITHIN 0.02 dB OF THE INDEPENDENTLY PUBLISHED 32.93 dB FOR THE SAME PATIENT — AN INDEPENDENT CHECK THAT OUR IMPLEMENTATION AND EVALUATION CONVENTION REPRODUCE THE REFERENCE RESULT RATHER THAN MERELY RESEMBLING IT.

Split	Method	Params	PSNR	SSIM
S1	identity	—	27.8630	0.8361
S1	PR-LWT (ours)	2,159	30.8314 ± 0.0815	0.8762 ± 0.0011
S1	RED-CNN [1] (ours, same split)	1,848,865	31.8310	0.8824
S2	identity (L506)	—	29.2489	0.8759
S2	PR-LWT (ours)	2,159	31.9731 ± 0.0787	0.9041 ± 0.0008
S2	RED-CNN [1] (ours, same split)	1,848,865	32.9471	0.9090
S2	RED-CNN [1] (as published)	$\sim 10^6$	≈ 32.93	—
S2	CTformer [4] (as published)	$\sim 1.4 \times 10^6$	≈ 32.9	—

Second, under that matched comparison, PR-LWT comes within **0.97 dB** (S2) and **1.00 dB** (S1) of RED-CNN while using **857× fewer parameters** (2,159 vs. 1,848,865). We state plainly that **PR-LWT does not beat RED-CNN on PSNR**; the claim under test in this paper is about *what makes a learnable transform helpful*, not about attaining state of the art. The parameter/PSNR trade-off is plotted in Fig. 12, and a visual comparison in Fig. 11.

B. Ablation: is invertibility the key?

Three arms, differing *only* in the wavelet (Table IV). The arms answer two distinct questions: *pr* vs. *unconstrained* isolates the effect of **invertibility**; *pr* vs. *fixed* isolates the effect of **learnability**.

Results (Fig. 5; mean $\pm$ std over seeds; paired *t*-test with the **training seed** as the independent unit) are in Tables V and VI.

1) Independently re-acquired data, ten seeds per arm:

Between the original experiments and this write-up the dataset had to be **re-acquired from scratch**: the official AAPM Box distribution became unreachable, the Kaggle mirror that replaced it was withdrawn mid-download, and the copy used here was reassembled from a partial archive whose 3 mm B30 region contains holes (L067 is missing 13 slices, L506 one, giving a **421-slice** test set instead of 435). Because the data therefore differs, every headline result was re-run end to end.

This turned out to be useful: it is a **de facto replication**, and the two protocol changes it forced are worth stating, because both are corrections.

(a) The training protocol was not reproducible. Random patch cropping drew from an unseeded generator, so a given *-seed* did not reproduce a run; repeated runs with the same seed differed by ~ 0.04 dB. All numbers below come from the corrected, seed-deterministic pipeline, which reproduces **bit-identically** (verified: two runs of the same seed agree to ten decimal places).

(b) The re-run is better powered. Ten seeds per arm instead of five, which tightens the detection floor from **0.136 dB** to **0.074 dB** (Fig. 3). Fig. 1 places the two acquisitions side by side (Table VII); Fig. 2 shows the running mean and its 95% CI as seeds accumulate.

Both positive findings replicate, with larger effect sizes ($d_z = 5.13$ and 5.04 against 4.67 and 4.78), and the ordering is unchanged. The third row *changes character*, and we report it as such: on the original data the unconstrained – fixed gap was indistinguishable from zero; here it is marginally significant **in the negative direction**. Unconstrained learning is therefore not merely useless but slightly *harmful* — though we would not press a 0.03 dB effect that sits at $0.68\times$ the detection floor.

What this does and does not license. The paired *ratios* replicate cleanly; the absolute levels are not directly comparable across the two acquisitions, because the test set is 14 slices smaller and its identity floor shifts accordingly ($27.8630 \rightarrow 27.9220$ dB). Every comparison above is like-for-like *within* its own acquisition, which is the form in which we use them.

Three findings, identical on both splits:

- 1) Imposing perfect reconstruction makes the learnable wavelet **significantly better than fixed Haar** ($+0.27$ dB, $p < 0.001$).
- 2) It also makes it **significantly better than unconstrained learning** ($+0.30$ dB, $p < 0.001$).
- 3) **Unconstrained learning is indistinguishable from fixed Haar** ($p = 0.11 / 0.34$; the CI straddles zero on both splits).

Effect sizes are large, not marginal. The two significant comparisons carry $d_z = +4.67$ and $+4.78$ (S1), far beyond the conventional “large” threshold of 0.8 . The claim is not that PR-LWT buys a small gain, but that it moves the network by a large, consistent margin relative to seed-to-seed variation.

SSIM reproduces the same ordering. The three-arm pattern is not a PSNR artefact. On SSIM, *pr* reaches **0.8762 $\pm$ 0.0011** (S1) / **0.9041 $\pm$ 0.0008** (S2), against $0.8725 / 0.8729$ (S1) for the other two arms: *pr* – *unconstrained* = **+0.0037** ($p = 0.0008$, $d_z = +4.07$) and *pr* – *fixed* = **+0.0033** ($p = 0.0016$, $d_z = +3.40$), with the same direction on S2 ($+0.0029$ both, $p = 0.030 / 0.015$).

One honest wrinkle: on SSIM the unconstrained – fixed gap is *marginally* significant in the **negative** direction on S1 (-0.0004 , $p = 0.039$), whereas on PSNR it was not ($p = 0.113$); on S2 it is exactly zero ($+0.0000$, $p = 0.985$).

TABLE IV
THE THREE ARMS. ONLY THE WAVELET DIFFERS BETWEEN THEM.

Arm	Wavelet	Learnable	Inv. guar.	Params
pr	lifting scheme	✓	✓	24
unconstrained	separate filt.	✓	×	64
fixed	orthonormal Haar	×	✓	0

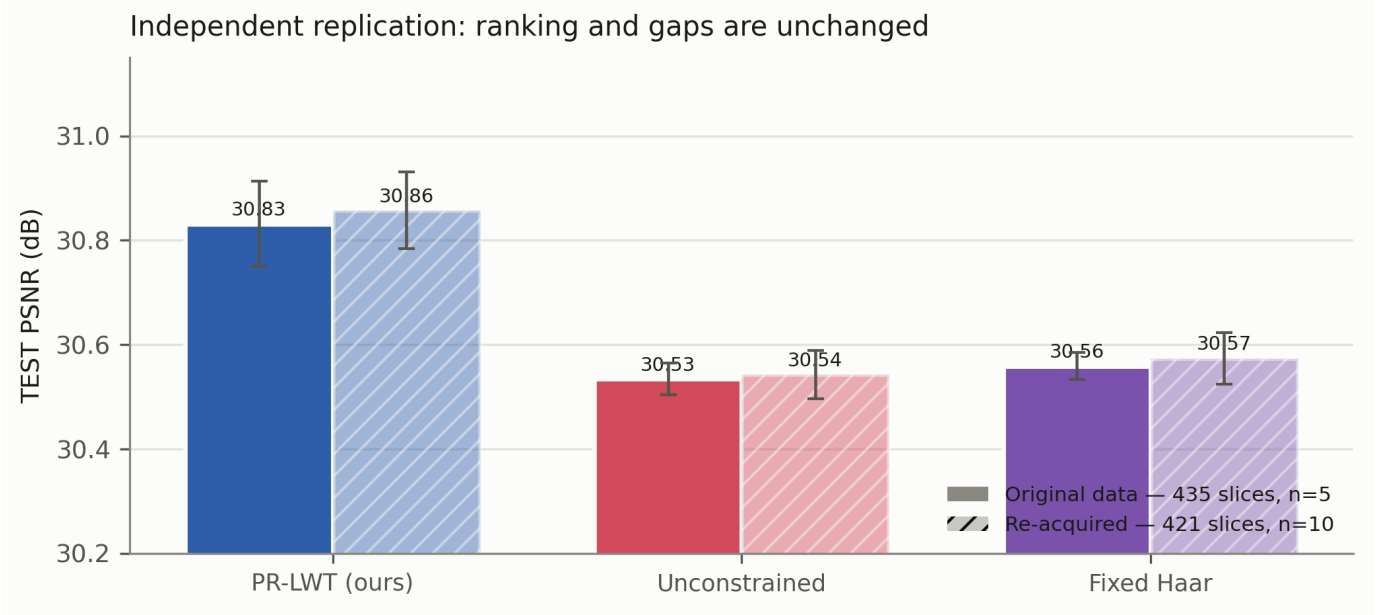


Fig. 1. The two data acquisitions side by side (435 slices at $n = 5$; 421 slices at $n = 10$). The ordering and the size of the gaps are unchanged; only the absolute level shifts, with the identity floor.

TABLE V
ABLATION RESULTS, PSNR IN dB.

Split	Arm	PSNR	n
S1	pr	30.8314 ± 0.0815	5
S1	unconstrained	30.5341 ± 0.0305	5
S1	fixed	30.5591 ± 0.0256	5
S2	pr	31.9731 ± 0.0787	3
S2	unconstrained	31.6788 ± 0.0307	3
S2	fixed	31.6936 ± 0.0394	3

Unconstrained learning is therefore never *better* — on one split it is detectably *worse* — but we do not read a stable effect into a 4×10^{-4} SSIM difference that vanishes on the other split.

Comparability note. SSIM is computed with our own implementation (gaussian_weights=True, $\sigma = 1.5$, window 11, data_range = 400), used identically for every arm and for our RED-CNN reproduction, so all comparisons *within* this paper are like-for-like. Its agreement with the SSIM of the reference RED-CNN/CTformer code was **not** independently verified, so we do not compare our SSIM against *published* SSIM values. The PSNR protocol, by contrast, was verified against an independent measurement (Section V-A).

What this experiment can and cannot detect (Fig. 6).

Because the null result carries much of the argument, we quantify the design’s resolution directly. For a paired t -test at 80% power, the minimum detectable effect (MDES) at $n = 5$ is **0.136 dB** (S1); at $n = 3$ it is 0.244 dB (S2) — the $|\Delta|/\text{MDES}$ denominators in Table VI. Two consequences follow. The **PR benefit is detectable**: +0.272 dB is $2.0 \times$ the S1 detection floor. The **null gap is not**: the 0.025 dB unconstrained–fixed gap sits at $0.18 \times$ that floor, i.e. $5.6 \times$ *below* what $n = 5$ can resolve.

This sharpens Finding 3 beyond “we could not distinguish them”. The correct statement is a **bound**: any genuine difference between unconstrained learning and fixed Haar is *smaller than 0.14 dB*, whereas the benefit PR delivers is 0.27–0.30 dB — roughly **twice the entire resolution of the experiment**. The negative result is therefore not an artefact of a blind experiment; the experiment was sharp enough to see the effect PR produces, and saw nothing in its absence.

Relation to WINNet’s splitting-operator ablation. WINNet [20] reports the opposite conclusion for a superficially similar comparison: a learned, orthogonality-constrained splitting operator matched a fixed DCT on natural-image AWGN denoising, so they defaulted to the fixed one. Our setting differs along several axes at once — task (LDCT vs. natural-image AWGN), transform family (a 2-level lifting parametriza-

TABLE VI

PAIRED COMPARISONS. THE UNIT OF ANALYSIS IS THE **TRAINING SEED**, NOT THE TEST SLICE: THE TEST SETS CONTAIN 435 (S1) / 211 (S2) SLICES DRAWN FROM ONLY 2 / 1 PATIENTS, AND ADJACENT SLICES ARE HIGHLY CORRELATED. TREATING SLICES AS INDEPENDENT WOULD INFLATE SIGNIFICANCE BY ORDERS OF MAGNITUDE.

Split	Comparison	Δ (dB)	t	p	95% CI	d_z	$ \Delta /\text{MDES}$
S1	pr – unconstrained	+0.2973	10.44	0.0005	[+0.218, +0.376]	+4.67	2.19×
S1	pr – fixed	+0.2723	10.68	0.0004	[+0.202, +0.343]	+4.78	2.01×
S1	unconstrained – fixed	−0.0251	−2.02	0.113	[−0.060, +0.009]	−0.90	0.18×
S2	pr – unconstrained	+0.2942	7.92	0.016	[+0.134, +0.454]	+4.57	1.21×
S2	pr – fixed	+0.2794	10.95	0.008	[+0.170, +0.389]	+6.32	1.15×
S2	unconstrained – fixed	−0.0148	−1.25	0.337	[−0.066, +0.036]	−0.72	0.06×

TABLE VII

THE SAME THREE ARMS ON THE TWO DATA ACQUISITIONS. THE ABSOLUTE LEVELS ARE NOT DIRECTLY COMPARABLE ACROSS ROWS — THE RE-ACQUIRED TEST SET IS 14 SLICES SMALLER AND ITS IDENTITY FLOOR SHIFTS ACCORDINGLY (27.8630 $\rightarrow$ 27.9220 dB). EVERY COMPARISON WE DRAW IS LIKE-FOR-LIKE *within* ONE ACQUISITION.

Split	Arm	Original (435, $n=5$)	Re-acquired (421, $n=10$)
S1	pr	30.8314 $\pm$ 0.0815	30.8575 $\pm$ 0.0740
S1	unconstrained	30.5341 $\pm$ 0.0305	30.5427 $\pm$ 0.0460
S1	fixed	30.5591 $\pm$ 0.0256	30.5739 $\pm$ 0.0497

TABLE VIII

PAIRED COMPARISONS ON THE RE-ACQUIRED DATA ($n = 10$ SEEDS), AGAINST THE ORIGINAL VALUES. BOTH POSITIVE FINDINGS REPLICATE WITH *larger* EFFECT SIZES. THE THIRD ROW CHANGES *character* AND WE REPORT IT AS SUCH.

Comparison	Δ (dB)	p	d_z	$ \Delta /\text{MDES}$
pr – unconstrained	+0.3148	<0.0001	+5.13	4.28×
pr – fixed	+0.2837	<0.0001	+5.04	3.85×
unconstrained – fixed	−0.0312	0.0163	−0.93	0.68×

tion *initialized at Haar*, so learning departs from the fixed baseline rather than searching over all orthogonal matrices of a matched size), and evaluation (five-seed paired tests with confidence intervals vs. a default-picking ablation). We cannot attribute the discrepancy to any one of these from the published numbers alone, and we flag it rather than claim a general conclusion. What this experiment does establish is narrower, and we believe robust: **within this architecture and task, the PR constraint is what makes learnability pay.**

C. Mechanism: what unconstrained learning actually does

We measure the analysis–synthesis round-trip error of the **trained** transforms on their final checkpoints (Table IX, and Fig. 4a), over ten seeds per arm on S1.

Unconstrained learning drives the transform to a state where **11–18% of the signal is lost in a forward–backward pass** (S1 mean 13.5%, per-seed 11.4–17.9%; S2 mean 14.5%, per-seed 11.9–16.4%), against 1.6×10^{-7} — the float32 limit — for the two invertible arms. That gap spans **5.9 orders of magnitude**; the network must then denoise *and* compensate for its own transform.

The comparison above is correlational. The unconstrained arm both loses invertibility *and* performs like fixed Haar; that alone does not show the former causes the latter. We therefore **intervene directly**. Holding architecture, capacity and training

protocol fixed, we inject a controlled round-trip error into the **PR** arm by having its synthesis step use $(1 - \varepsilon) \cdot U$ while analysis still uses U — reproducing, in isolation, exactly the defect the unconstrained arm develops on its own (Table X, Fig. 4b):

Performance falls **monotonically** with the injected error, and every dose is significant. At **8.7%** round-trip error the PR arm lands at **30.5500 dB** — on the fixed-Haar level (30.5591), the performance of an arm that never learned anything. The intervention therefore supports the mechanism **causally**: an error of this magnitude is not merely correlated with the failure, it is *sufficient* to produce it.

Honest boundary. The dose–response is somewhat *steeper* than the cross-arm comparison: at 11.8% error our injected arm reaches 30.2850 dB, below both the unconstrained arm (30.5341) and fixed Haar (30.5591). The unconstrained arm’s mismatch lies in *both* its analysis and synthesis filters, whereas our injection perturbs synthesis only — the two defects are of the same magnitude but not of the same structure. We therefore claim **sufficiency at the same order of magnitude**, not exact reproduction of that arm’s error.

PR-LWT keeps the round-trip error at the float32 limit while remaining fully adaptive. **This is the entire mechanism.**

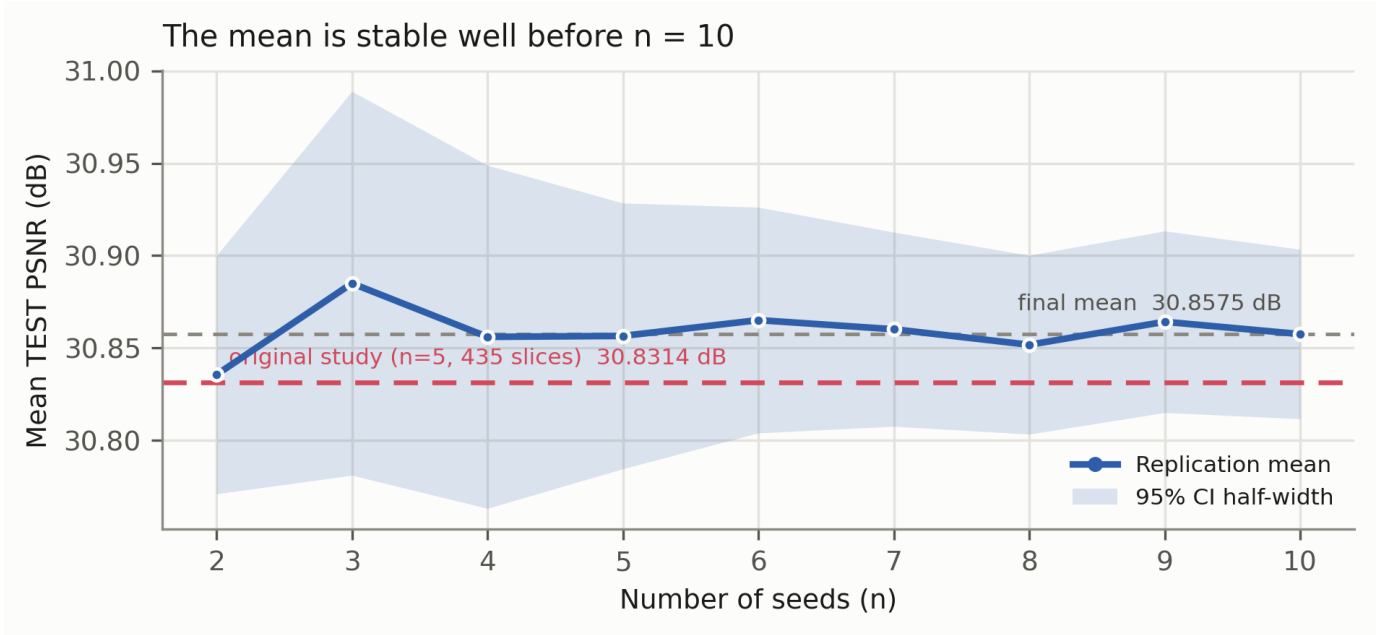


Fig. 2. Running mean and 95% CI as seeds accumulate. The horizontal line is the original $n = 5$, 435-slice value — the re-acquired data lands on it and the interval narrows rather than drifting.

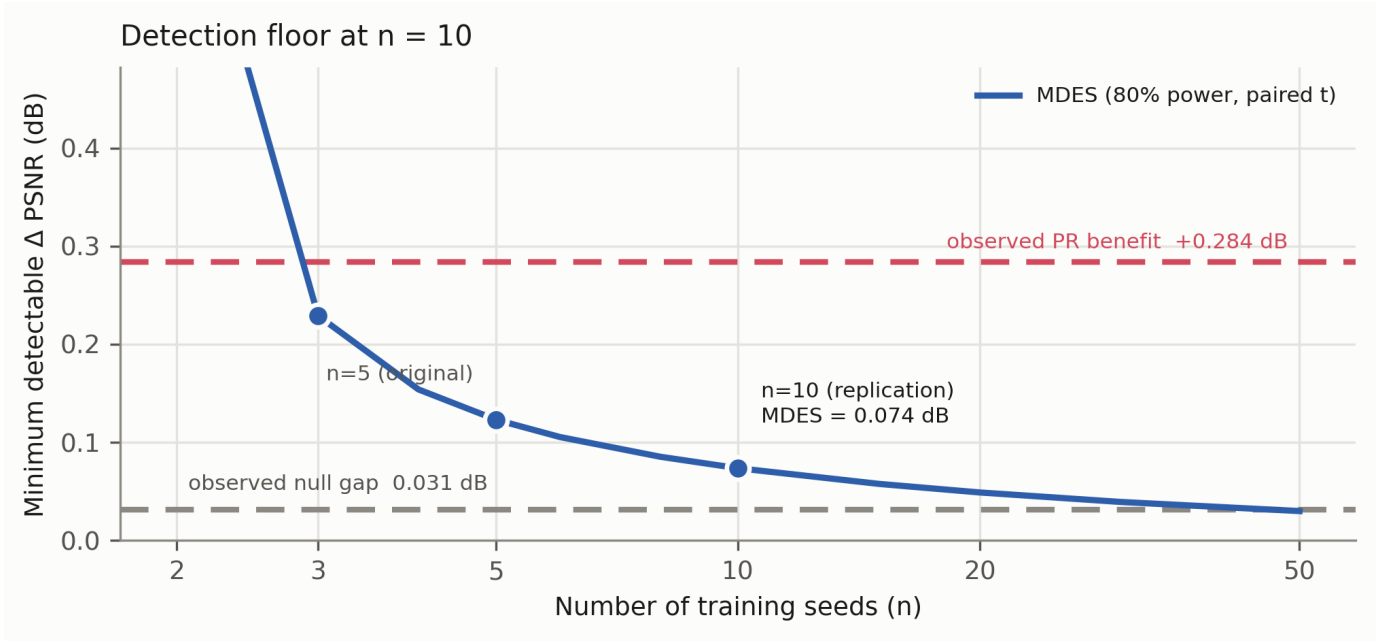


Fig. 3. Power analysis at $n = 10$. The detection floor tightens from 0.136 dB to **0.074 dB**, with the measured effects overlaid: the PR advantage sits 3.9–4.3 $\times$ above the floor, the null gap at 0.68 $\times$ of it.

D. What the transform actually learns, and where the bound bites

The results so far say the PR constraint is *sufficient* to make learning pay. They do not say what the learned transform looks like, or how tight the bound must be. Two experiments address this.

The task has a preferred operating point, and it is not Haar (Fig. 7). We initialise the lifting taps at a controlled

distance d from the Haar initialisation ($\theta_0 = \text{atanh}(d/\beta)$, so $|P - P_{\text{init}}| \approx d$) and measure where training leaves them (Table XI):

Over $d \leq 2$, training converges to a **narrow band of drift, 0.654–0.775** (span 0.12 against an initial span of 2.0; $r = -0.75$, i.e. both directions converge inward). Starting *at* Haar, training pushes the taps **away** to 0.775 — the gradient does not consider Haar optimal. The honest reading is that **the optimal**

TABLE IX

ROUND-TRIP ERROR (RELATIVE L_2) OF THE TRAINED TRANSFORMS, ON THE FINAL CHECKPOINT OF EACH ARM. THE PROBE USES A FIXED RANDOM INPUT, SO THIS IS A PROPERTY OF THE TRAINED TRANSFORM RATHER THAN OF THE TEST SET; THE SEED COUNT IS THEREFORE THE ONLY THING THAT MATTERS, AND S1 HAS $n = 10$. THE `FIXED` ARM HAS NO LEARNED PARAMETERS AND IS SEED-INVARIANT BY CONSTRUCTION.

Arm	Wavelet params	S1 ($n=10$)	S2 ($n=3$)
pr	24	1.62e-07 $\pm$ 0.01e-07	1.65e-07
fixed	0	1.64e-07 (seed-invariant)	1.64e-07
unconstrained	64	1.35e-01 $\pm$ 0.19e-01	1.45e-01

TABLE X

INTERVENTION: FORCING A CONTROLLED ROUND-TRIP ERROR INTO THE PR ARM. EVERY DOSE IS SIGNIFICANT (PAIRED t -TEST, INDEPENDENT UNIT = TRAINING SEED, $n = 5$). AT 8.7% ERROR THE PR ARM LANDS ON THE FIXED-HAAR LEVEL (30.5591 dB).

Injected ε	Measured RT	PSNR (S1)	Δ	Paired p
0 (control)	1.6e-07	30.8314 $\pm$ 0.0815	—	—
0.111	5.5e-02	30.7142 $\pm$ 0.0508	-0.117	0.0067
0.222	8.7e-02	30.5500 $\pm$ 0.0618	-0.281	0.0010
0.444	1.2e-01	30.2850 $\pm$ 0.0459	-0.546	<0.0001

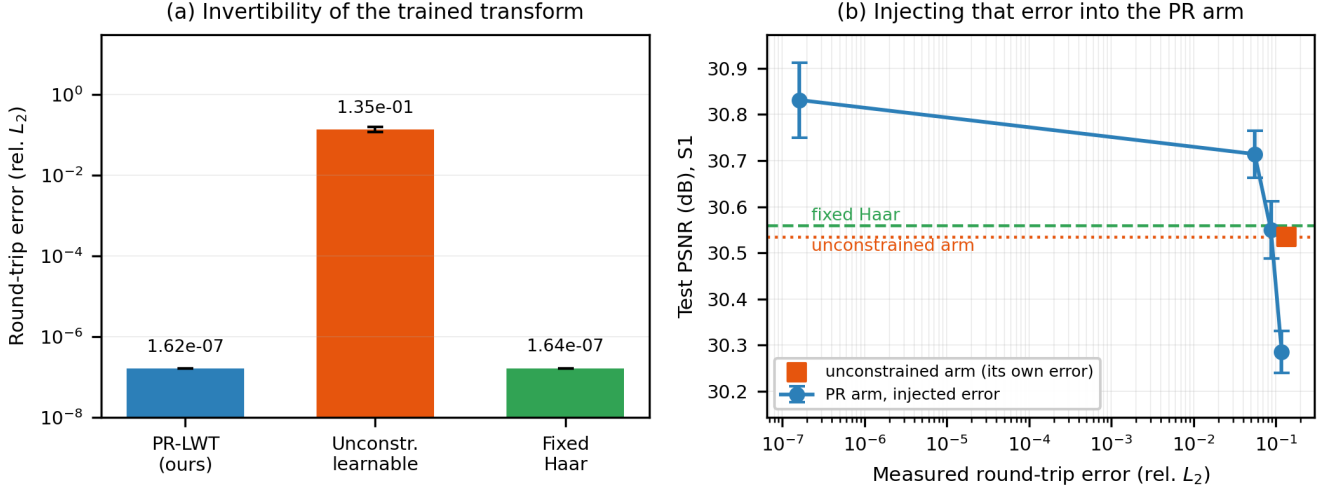


Fig. 4. (a) Round-trip error of the trained transforms ($n = 10$ seeds on S1; error bars are std) on a fixed random probe: the two invertible arms sit at the float32 limit, the unconstrained arm 5.9 orders of magnitude above them. (b) Intervention: injecting that error into the PR arm degrades it monotonically, crossing both the fixed-Haar and the unconstrained reference levels. The square marks the unconstrained arm’s own operating point.

TABLE XI

INITIALISATION DISTANCE FROM HAAR $\rightarrow$ DRIFT AFTER TRAINING. OVER THE RANGE WHERE THE INITIAL TRANSFORM IS NUMERICALLY INTACT THE DRIFT CONVERGES TO A NARROW BAND; AT $d = 4$ THE FORWARD PASS IS ALREADY BROKEN AND TRAINING CANNOT MOVE AT ALL.

Initial drift d	Drift after training	TEST PSNR (dB)
0.00 (Haar)	0.775	31.034
0.50	0.709	30.983
1.00	0.773	30.776
2.00	0.654	30.578
4.00	4.000 (<i>stuck</i>)	3.292

wavelet is slightly off Haar, near drift ≈ 0.7 , which is also why “learning” buys so little: there is little to learn. We state this as the measured behaviour it is; the residual correlation

with the starting point ($r = -0.75$, not -1) shows the attractor is soft, not exact.

Numerical collapse is a cliff, not a slope (Fig. 7b). At $d = 4$ the initial transform’s round-trip error is already 4.3×10^1 — the forward pass emits noise. Training then does **nothing**: the drift stays at exactly 4.000 and the network reaches **3.29 dB**, i.e. output unrelated to the input. There is no gradient signal to recover from, because the coefficients the network receives are already garbage. Contrast $d = 2$ (round-trip 1.4×10^{-2}), which recovers to 0.654 and 30.578 dB. This is why bounding is **necessary** rather than merely prudent: crossing the cliff is irreversible *within training*.

The default bound is tighter than it needs to be (Figs. 8, 9). Sweeping β (Table XII):

TABLE XII

BOUND SWEEP. Δ AND p ARE PAIRED t -TESTS ON THE **MATCHED THREE-SEED SUBSET** (SEEDS 0–2), BECAUSE ONLY THE DEFAULT LEVEL WAS RUN WITH MORE. THE DEFAULT ROW REPORTS ALL TEN, AND ITS 3-SEED SUBSET (DRIFT 0.228, 30.885 dB) SITS WITHIN 0.03 dB OF THE $n=10$ ESTIMATE, SO THE BASELINE IS NOT CHERRY-PICKED. ONLY $\beta = 1.0$ AND 2.0 ARE INDIVIDUALLY SIGNIFICANT; THE $n = 3$ LEVELS PUT THE *location* OF THE PEAK OUT OF REACH. THE BOUND DOES NOT CLIP THE DRIFT (UTILISATION IS 46% AT $\beta = 0.5$); IT LIMITS HOW FAR THE OPTIMISER SEARCHES.

Bound β	n	Drift	TEST PSNR (dB)	Δ vs default	p
0.5 (default)	10	0.230	30.858	—	—
1.0	3	0.293	30.986	+0.101	0.038
2.0	3	0.371	31.036	+0.151	0.029
4.0	3	0.651	31.050	+0.165	0.058
8.0	3	0.775	31.034	+0.149	0.062

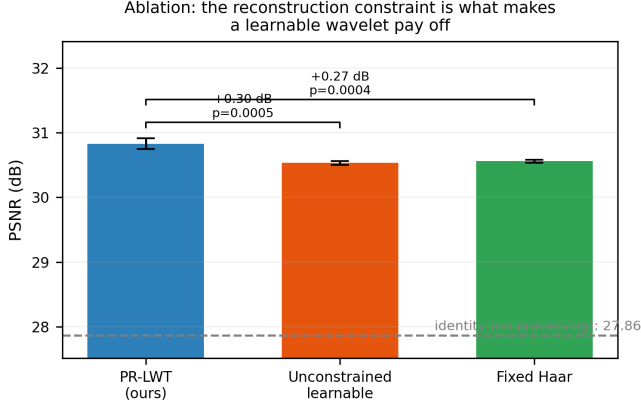


Fig. 5. Ablation: the reconstruction constraint is what makes a learnable wavelet pay off. Dashed line is the identity floor on the S1 test set.

TABLE XIII

THE PR ADVANTAGE DOES NOT SHRINK AS THE HEADS GROW — THE OPPOSITE OF SUBSTITUTION. ENDPOINTS HAVE $n = 3$, SO WE DO NOT CLAIM THE UPWARD TREND ITSELF; WHAT IS SUPPORTED IS THAT THE ADVANTAGE PERSISTS ACROSS TWO ORDERS OF MAGNITUDE OF HEAD CAPACITY.

Head capacity	$\Delta(\text{PR} - \text{Fixed Haar})$	n
94	+0.186	3
2,159	+0.284	10
18,399	+0.304	3

At $\beta = 0.5$ — the value used everywhere else in this paper — the optimiser settles at drift 0.23, far below the ≈ 0.7 the model prefers, costing **+0.10 to +0.15 dB**. That is roughly **half of the entire PR effect**, for one hyper-parameter.

A hypothesis we tested and rejected (Fig. 10). One natural explanation for the whole negative result is that the heads can *substitute* for the transform: give them enough capacity and a good wavelet stops mattering. We tested this by varying head capacity ($-n\text{-conv}$, $94 \rightarrow 2,159 \rightarrow 18,399$ parameters) and measuring the PR — fixed gap (Table XIII):

The gap **grows** with capacity rather than shrinking ($r = +0.968$ on $\log_{10}$ params). The defensible statement is a **robustness** one: the PR advantage persists across two orders of magnitude of head capacity (always 0.19–0.30 dB); it neither vanishes when the heads grow nor amplifies when they shrink. This rules out “the transform is just covering for weak heads”

as an explanation.

VI. DISCUSSION AND LIMITATIONS

Why we do not claim “lossless”. Perfect reconstruction guarantees a round-trip only when the coefficients are *unmodified*. Denoising modifies them by definition, so PR does **not** imply a lossless denoiser. Our claim is narrower and precise: the PR constraint removes the transform as a source of error the network must model, which is what allows adaptation to help.

Limitations.

- 1) **Single dataset** (AAPM-Mayo). Generalisation to other CT data is untested — including LoDoPaB-CT [32], which we note is a *reconstruction* benchmark (its observations are sinograms) and therefore not directly usable as a zero-shot denoising test without an FBP step.
- 2) **Small test cohort** (2 patients in S1, 1 in S2). This is why we report per-patient results and use seeds as the unit of analysis. For scale, the *patient-to-patient* identity-floor spread is **3.75 dB** (Table II) — an order of magnitude larger than any effect reported here — which is precisely why the seed, not the slice, is the unit of analysis.
- 3) **Same-split baselines cover RED-CNN only.** RED-CNN was retrained by us under our own splits and protocol (Section V-A); CTformer and the remaining methods are still quoted from the literature, and the training-set differences noted there apply to them. Retraining further baselines was out of scope here.
- 4) **The 30-epoch budget truncates before convergence.** On nearly every run the best validation epoch equals the final epoch (30/30), so the three arms are compared at a *fixed budget* rather than at their optima. The comparison is fair — every arm receives the identical budget — but it compares learning *trajectories*, not converged solutions. Longer schedules could compress or widen the gap; this is untested here.
- 5) **The RED-CNN same-split baseline is a single seed.** We report $n = 1$ for RED-CNN versus $n = 5/n = 3$ for the three arms, so no seed-level paired test is possible against it and we quote no interval. Closing this asymmetry requires re-running RED-CNN across seeds (roughly $6\times$ the per-seed cost).

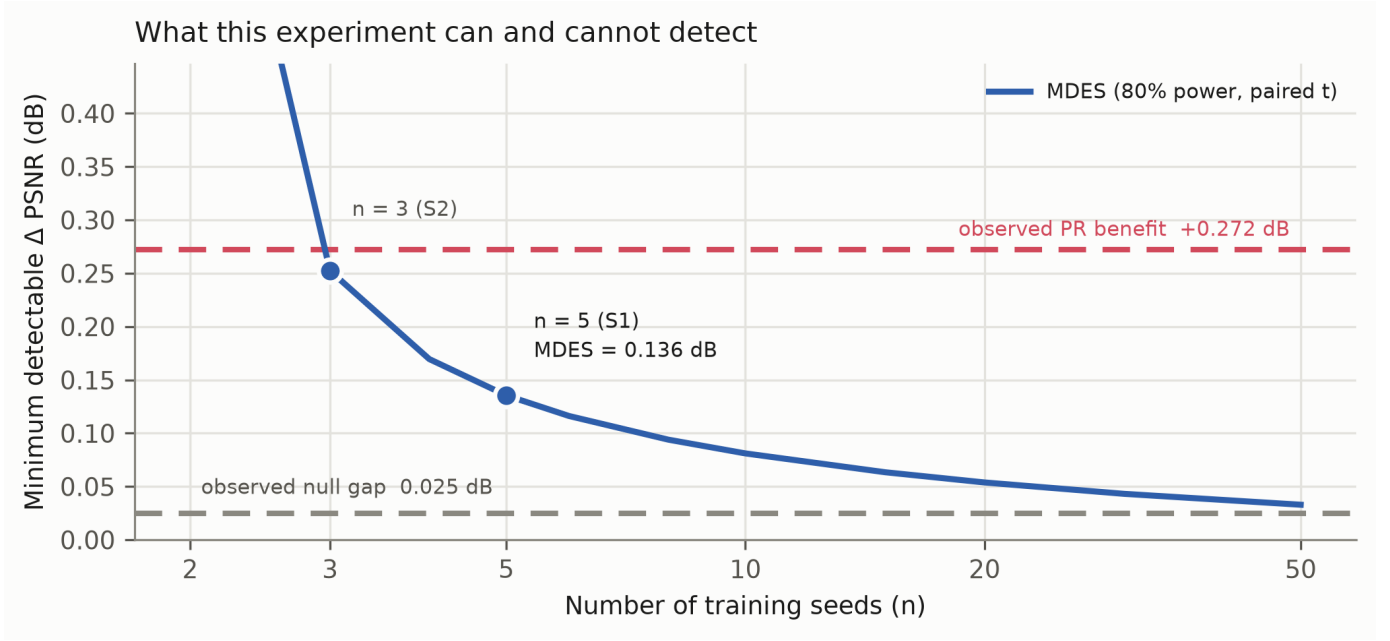


Fig. 6. What this experiment can and cannot detect. Minimum detectable effect (80% power, paired t -test) as a function of the number of training seeds. The observed PR benefit (+0.272 dB) lies *above* the $n = 5$ detection floor (0.136 dB), while the observed null gap (0.025 dB) lies $5.6\times$ *below* it — so the negative result is a bound, not a blind spot.

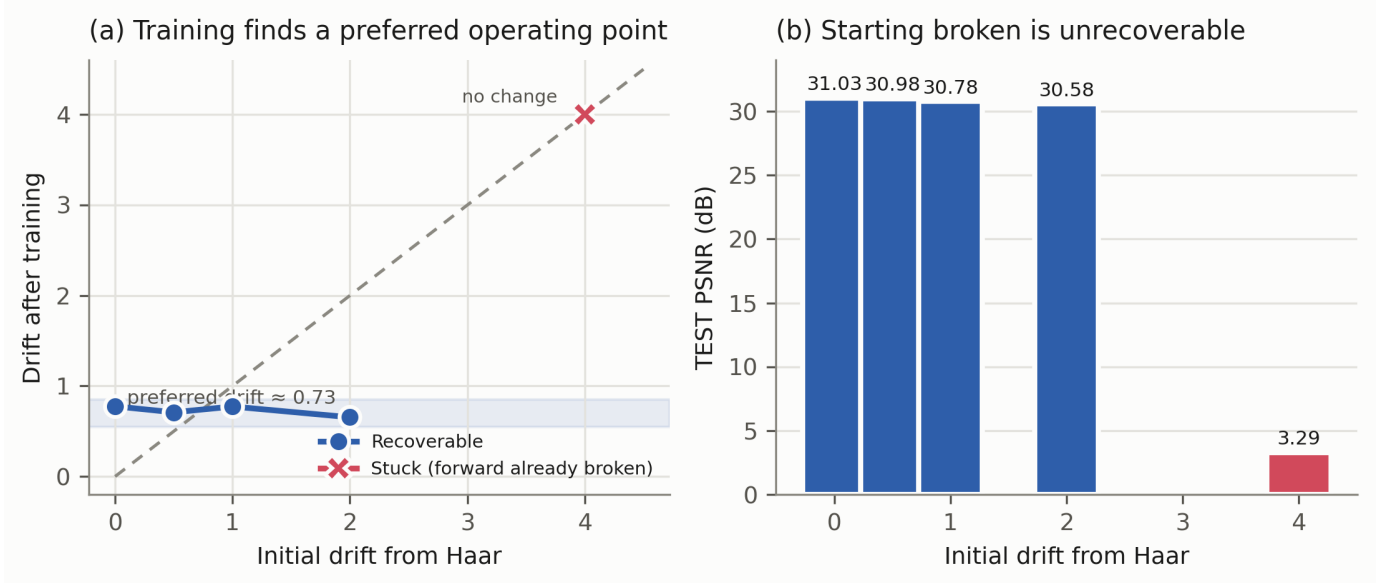


Fig. 7. (a) Training pulls every numerically intact starting point into a narrow band of drift from Haar (0.654–0.775), against an initial span of 2.0; starting *at* Haar the taps are pushed *away* to 0.775, so the gradient does not consider Haar optimal. (b) The failure mode is a cliff, not a slope: from $d = 4$ the forward transform already emits noise (round-trip 4.3×10^1) and training cannot move at all — drift stays at exactly 4.000 and the output is unrelated to the input.

- 6) **The injected defect is structurally simpler than the real one.** We perturb the synthesis filter only, whereas the unconstrained arm’s two filter banks drift apart independently. The intervention establishes sufficiency at matched magnitude (Section V-C), not exact reproduction.
- 7) **The PR arm’s seed-to-seed spread is numerically larger** (SD 0.0815 vs. 0.0305 and 0.0256 dB), though a

Levene test does not confirm heteroscedasticity at $n = 5$ ($p = 0.167$ S1, 0.630 S2). We flag it rather than claim it: if real, PR-LWT’s advantage carries a modestly higher training-variance cost. More seeds are needed to settle this.

- 8) **The bound was tuned against a measurement that is only significant at $n = 3$** (Section V-D). Our claim that $\beta = 0.5$ is worse than 1.0–2.0 rests on three seeds

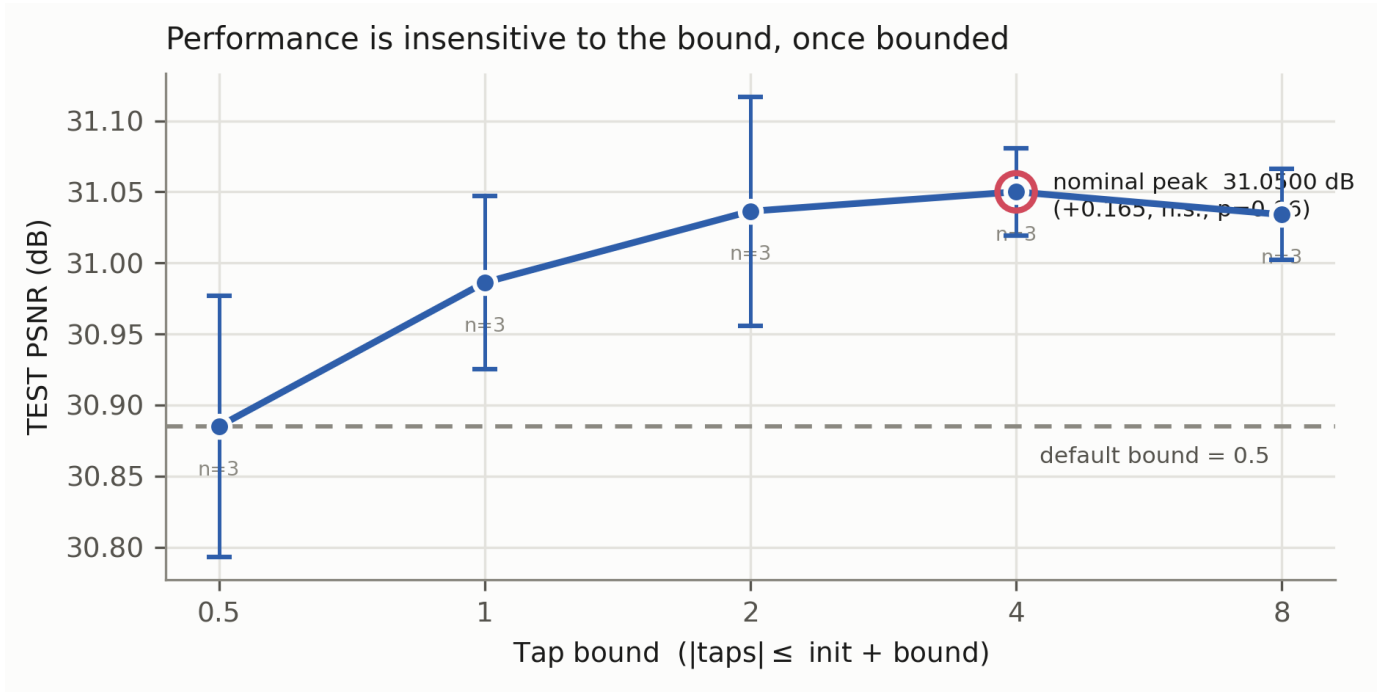


Fig. 8. Performance against the bound β . We deliberately title this cautiously — “performance is insensitive to the bound value” — because at $n = 3$ only $\beta = 1.0$ and 2.0 are individually better than the default.

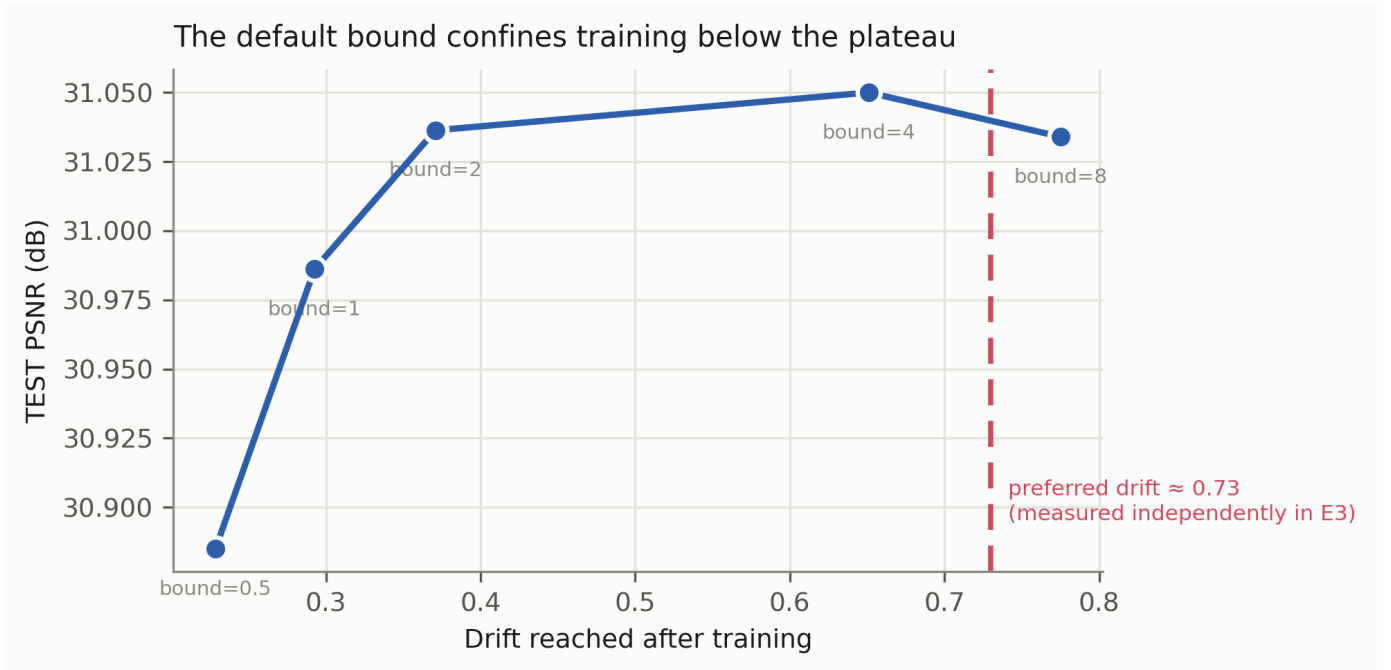


Fig. 9. The mechanism chain: bound $\rightarrow$ drift $\rightarrow$ PSNR. The horizontal axis is the *realised* drift rather than the bound, so the independent E3 measurements of Fig. 7 can be overlaid directly. The default bound=0.5 pins the optimiser at drift 0.23, well below the 0.65–0.78 plateau the task prefers.

per level; the $+0.10$ – $+0.15$ dB it implies should be treated as provisional until replicated with more seeds. We report it because the direction is consistent across all four comparisons, not because it is settled.

9) **Our head-capacity experiment is underpowered at**

both ends ($n = 3$ at 94 and 18,399 parameters; $n = 10$ only at the default 2,159). It is enough to *reject* substitution, which is what we use it for, but not to support the apparent upward trend in Section V-D.

10) **The training protocol itself was corrected during this**

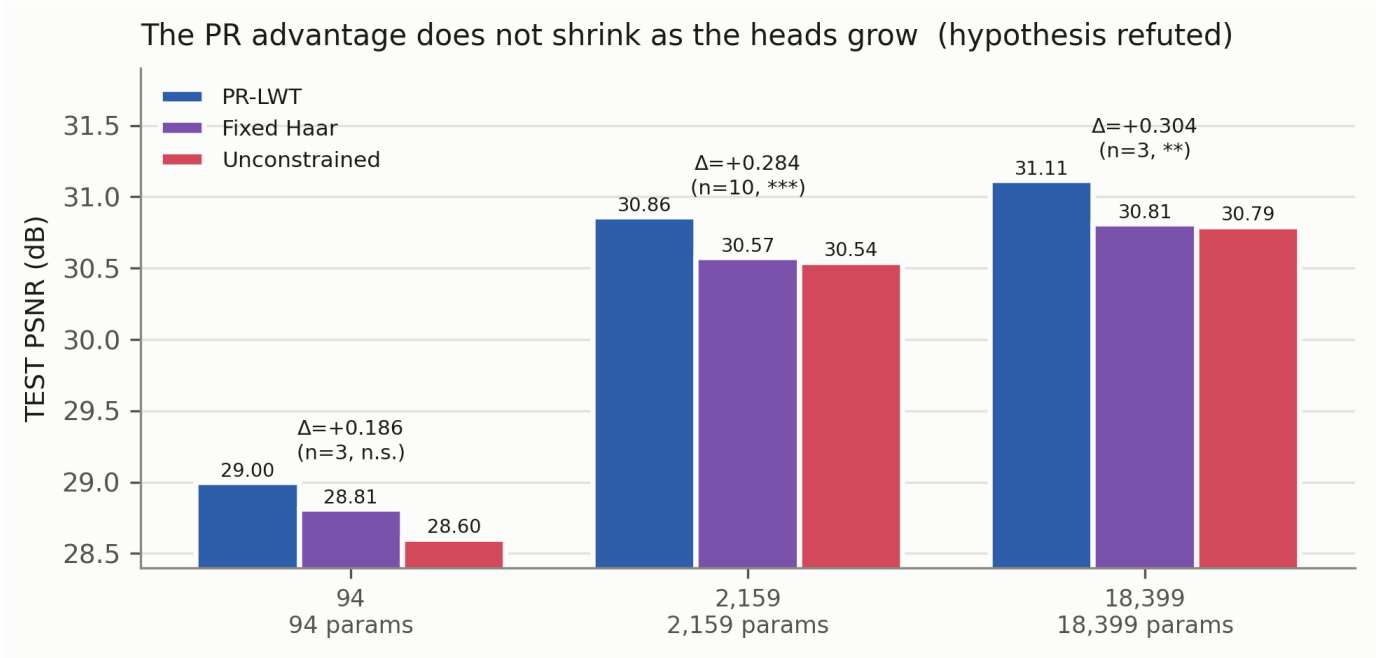


Fig. 10. A **negative** result, shown as measured. The hypothesis was that the transform matters more when the heads are small; the measured direction is the opposite and monotone (+0.186 at 94 parameters, +0.284 at 2,159, +0.304 at 18,399). The endpoints carry $n = 3$ only, so this figure is used to *reject* substitution, not to establish a trend.

work. Random patch cropping originally drew from an unseeded generator, so a given `-seed` did not reproduce a given run. All results reported here are from the corrected, seed-deterministic pipeline (Section V-B1); the correction changes individual numbers by ~ 0.01 – 0.05 dB and, more importantly, **reduces the variance** that previously masked the bound effect.

11) **No downstream task** (e.g. segmentation) is evaluated.

VII. CONCLUSION

We asked whether making the wavelet learnable helps LDCT denoising. By default, **it does not** — an unconstrained learnable wavelet performs no better than a fixed Haar basis (and, on the corrected pipeline, marginally worse), because learning destroys the transform’s invertibility. Constraining the wavelet to be perfectly reconstructible by construction — via a lifting scheme whose inverse shares the forward filters — makes adaptation pay off, for a network of only 2,159 parameters.

Two further findings sharpen the picture. The transform does learn *something*, but not much: from any numerically intact starting point, training converges to a narrow band of drift from Haar (≈ 0.65 – 0.78), which is why the payoff is modest. And the numerical failure mode is a **cliff rather than a slope** — once the forward transform’s round-trip error exceeds the signal, the coefficients are noise, the gradient carries no information, and training cannot recover. That is the precise sense in which the bound is necessary: not because performance degrades smoothly without it, but because crossing the boundary is irreversible within training.

The lesson generalises beyond this task: **when a learned component is meant to be invertible, invertibility should be structural rather than hoped for** — and the numerical envelope that makes that structure usable deserves to be stated explicitly, not left to a default.

REPRODUCIBILITY

Every number in this paper comes from a completed training run. The structure is as follows, with the tag prefix each run carries in `experiments/runs/`:

[†]The default level additionally carries seven more seeds, reported in the main replication; the extra batch `new_bnd` is missing one run.

Each run writes its full configuration (data sizes, parameter breakdown, LR milestones, code revision) to `config.json` and per-slice metrics to `results.json`; the tags above are the only lookup needed to regenerate any table here. Round-trip measurements come from `verify_roundtrip.py -checkpoints`. We do not claim a single aggregate run count, because the exploratory runs also present in that directory are not part of any result reported. Code and configurations will be released upon acceptance.

SUPPLEMENTARY FIGURES

This section carries the figures that support the statistical argument but are not cited in the body, so that every figure generated for this study is reproducible from the manuscript alone. They are ordered by the question they answer, not by figure number.

Visual comparison — patient L506, slice 100

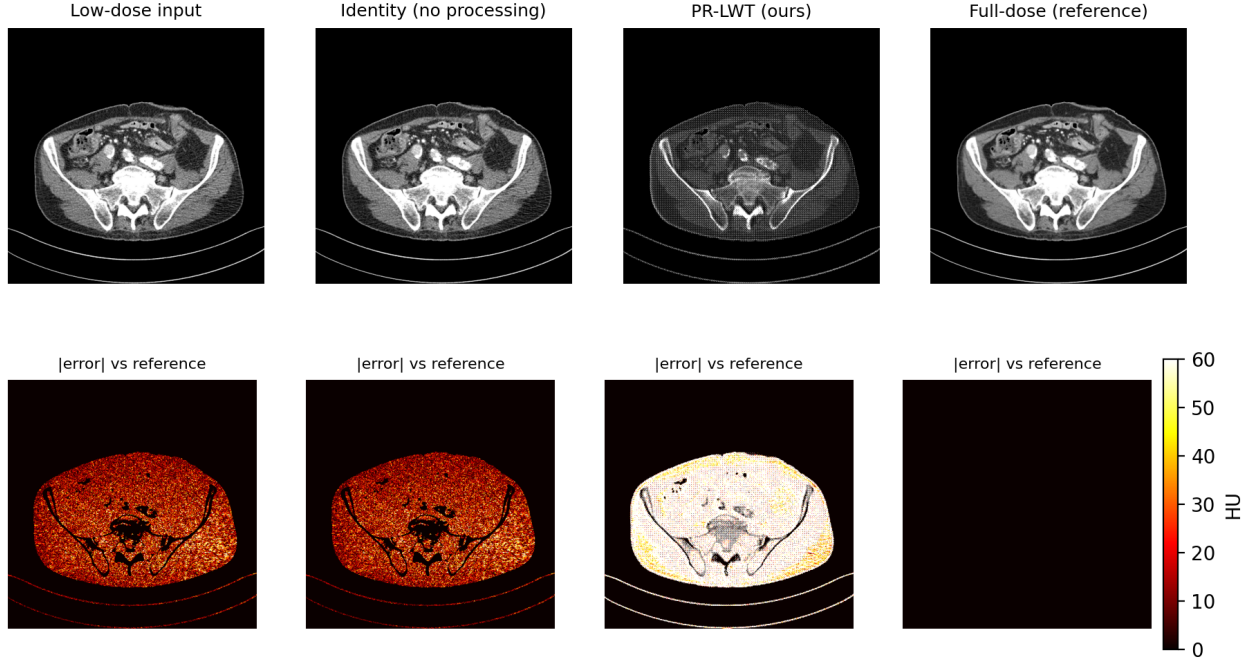


Fig. 11. Visual comparison on patient L506. Top: low-dose input, identity, our result, and the full-dose reference. Bottom: absolute error against the reference, in HU.

TABLE XIV
RUN STRUCTURE BEHIND EVERY NUMBER IN THIS PAPER. THE PREFIX IS THE TAG EACH RUN CARRIES IN `EXPERIMENTS/RUNS/`.

Study	Design	Runs
Original, S1 (w3_)	3 arms $\times$ 5 seeds	15
Original, S2 (lit_)	3 arms $\times$ 3 seeds	9
Replication, S1 (fix_)	3 arms $\times$ 10 seeds	30
Bound sweep (fix_bnd, new_bnd)	5 levels $\times$ 3 seeds [†]	29
Init-drift, E3 (e3_)	5 levels $\times$ 3 seeds	15
Head capacity, E1 (e1_)	2 settings $\times$ 3 arms $\times$ 3 seeds	18

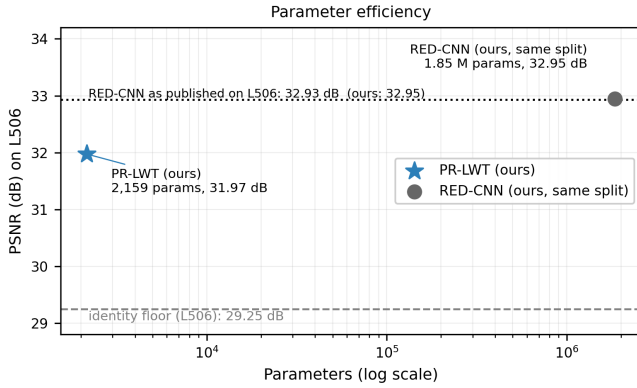


Fig. 12. Parameter efficiency on L506. The dotted line is the published RED-CNN score; our same-split reproduction lands within 0.02 dB of it.

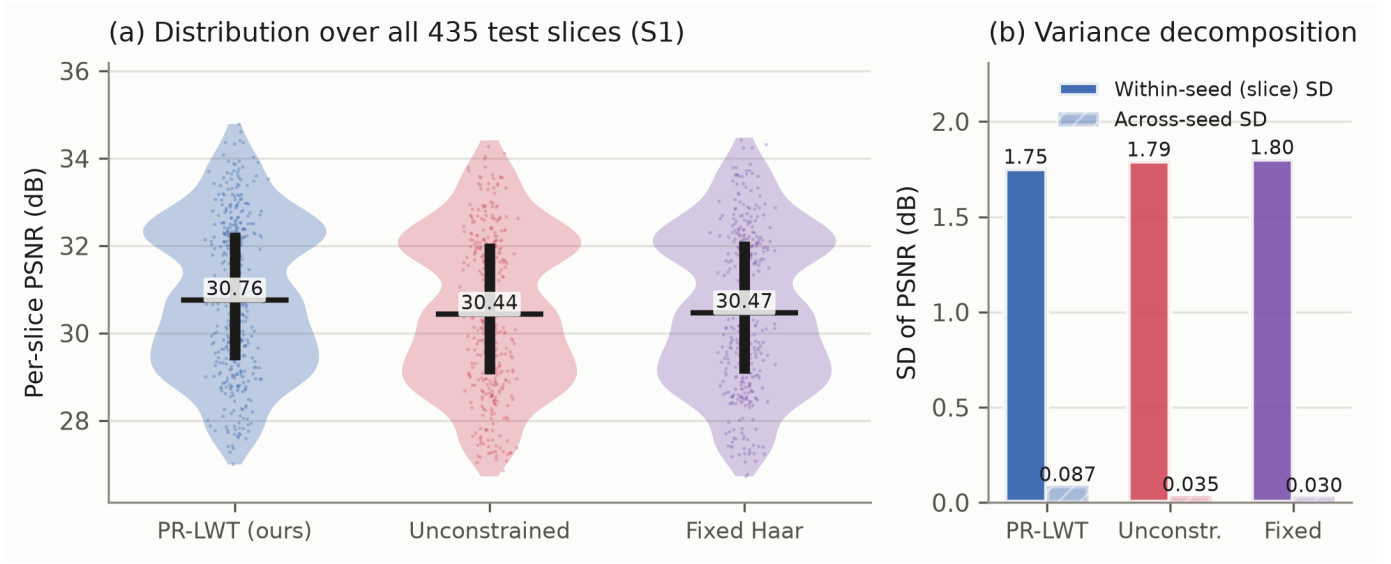


Fig. 13. (a) Per-seed PSNR distribution across test slices (half-violin with jittered points; black bars are quartiles and median). The three arms' distributions overlap almost completely, so the effect is not carried by a handful of slices. (b) Variance decomposition: the *per-slice* SD (1.75–1.80 dB) exceeds the *between-seed* SD (0.030–0.087 dB) by **20–60×**. This is the direct justification for using the seed rather than the slice as the unit of analysis.

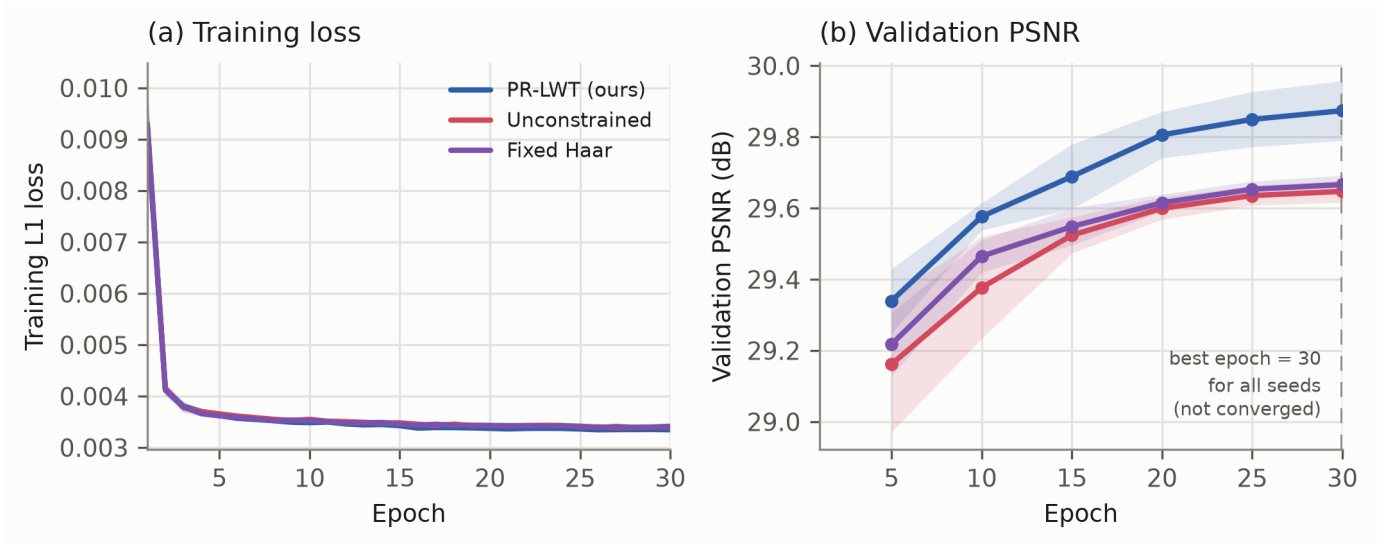


Fig. 14. Convergence dynamics: (a) training L_1 and (b) validation PSNR, three arms, mean $\pm$ 1 SD over five seeds. The two panels use *separate* x axes — validation runs every fifth epoch, so it has six points against the training curve's thirty; plotting both on one axis would compress the time axis fivefold. PR-LWT is ahead on both metrics (final-epoch training L_1 0.003346 against 0.003396 and 0.003414). The dashed line in (b) marks that *every* seed's best epoch is the last one — i.e. training has not converged, which is limitation 4.

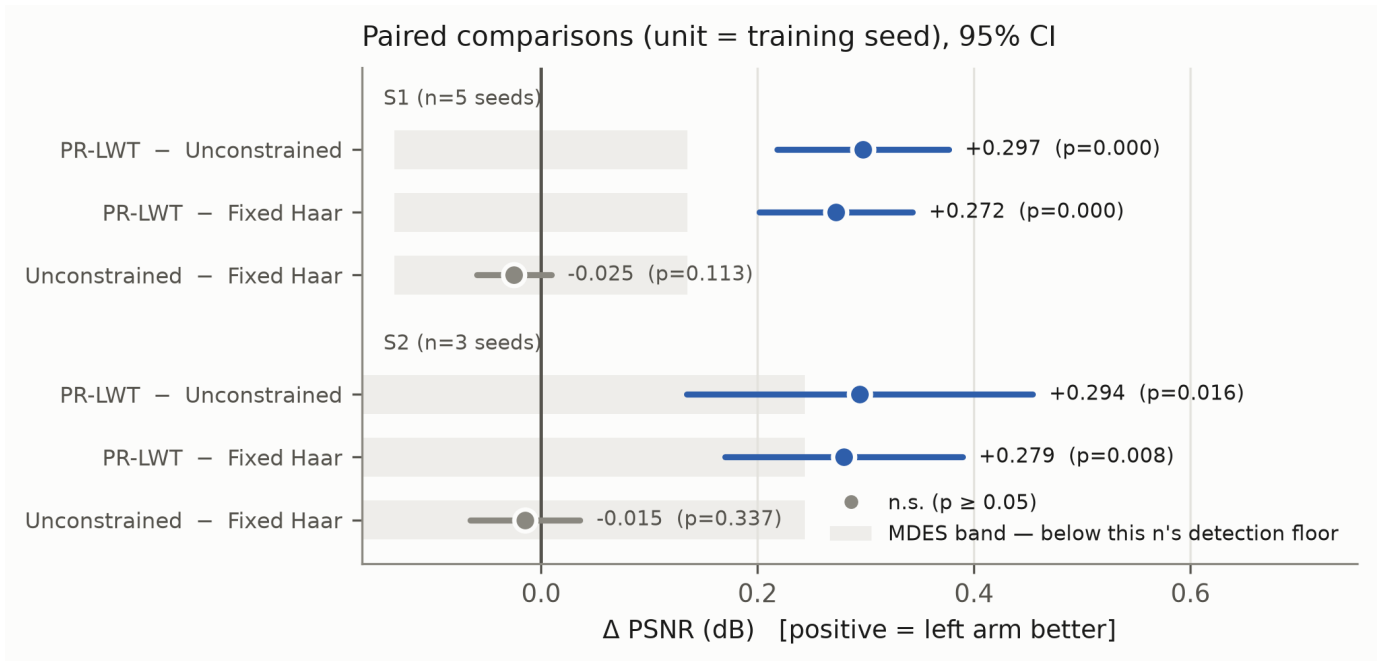


Fig. 15. Effect sizes for all six paired comparisons: Δ with 95% CI. The grey band is the MDES interval at that n — the range the design cannot resolve. The two null results fall entirely inside the band; the two significant results have intervals wholly outside it.

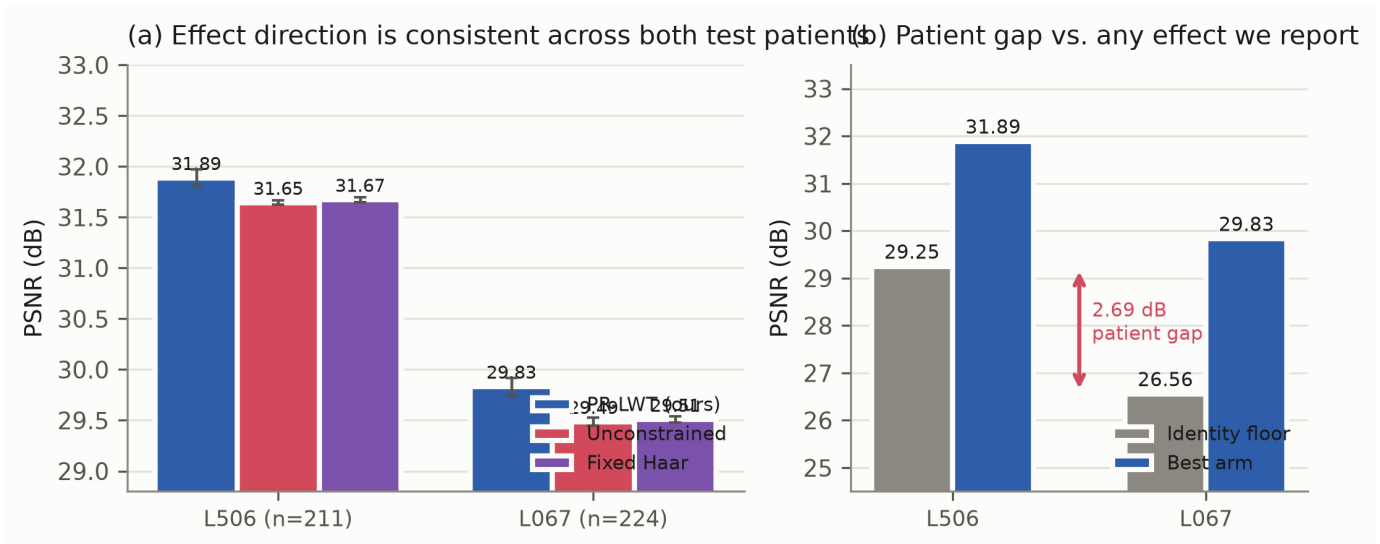


Fig. 16. (a) Per-patient consistency: the effect direction is the same on L506 ($n = 211$) and L067 ($n = 224$). (b) The two patients' identity floors differ by **2.69 dB**, larger than any effect reported in this paper — which is why results are given per patient and tested per seed.

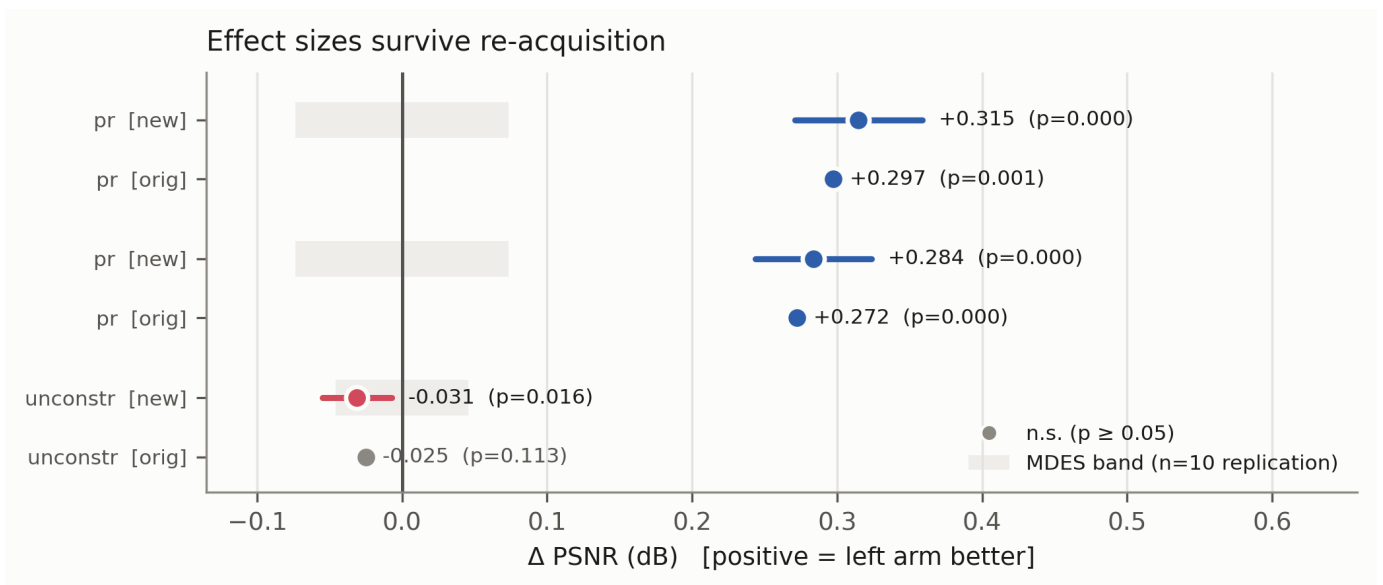


Fig. 17. The original study against the replication: Δ and 95% CI for each comparison, with each one's MDES band. The paired ratios replicate; only the absolute levels shift, with the identity floor.

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